

Patellofemoral Joint Loading during the Performance of the Wall Squat and Ball Squat with Heel-to-Wall-Distance Variations

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ABSTRACT

ESCAMILLA, R. F., N. ZHENG, T. D. MACLEOD, R. IMAMURA, K. E. WILK, S. WANG, K. YAMASHIRO, I. M. ESCAMILLA, and G. S. FLEISIG. Patellofemoral Joint Loading during the Performance of the Wall Squat and Ball Squat with Heel-to-Wall-Distance Variations. *Med. Sci. Sports Exerc.*, Vol. 55, No. 9, pp. 1592–1600, 2023. **Introduction:** Although bodyweight wall and ball squats are commonly used during patellofemoral rehabilitation, patellofemoral loading while performing these exercises is unknown, which makes it difficult for clinicians to know how to use these exercises in progressing a patient with patellofemoral pathology. Therefore, the purpose was to quantify patellofemoral force and stress between two bodyweight squat variations (ball squat vs wall squat) and between two heel-to-wall-distance (HTWD) variations (long HTWD vs short HTWD). **Methods:** Sixteen participants performed a dynamic ball squat and wall squat with long HTWD and short HTWD. Ground reaction force and kinematic data were used to measure resultant knee force and torque from inverse dynamics, whereas electromyographic data were used in a knee muscle model to predict resultant knee force and torque, and subsequently, all these data were inputted into a biomechanical computer optimization model to output patellofemoral joint force and stress at select knee angles. A repeated-measures two- and three-way ANOVA ($P < 0.01$) was used for statistical analyses. **Results:** Collapsed across long HTWD and short HTWD, patellofemoral joint force and stress were greater in ball squat than wall squat at 30° ($P = 0.009$), 40° ($P = 0.008$), 90° ($P = 0.003$), and 100° ($P = 0.005$) knee angles during the squat descent, and greater in wall squat than ball squat at 100° ($P < 0.001$), 90° ($P < 0.001$), 80° ($P = 0.004$), and 70° ($P = 0.009$) knee angles during squat ascent. Collapsed across ball and wall squats, patellofemoral joint force and stress were greater with a short HTWD than a long HTWD at 100° ($P = 0.007$) and 90° ($P = 0.008$) knee angles during squat ascent. **Conclusions:** Patellofemoral joint loading changed according to both squat type and HTWD variations. These differences occurred in part due to differences in forces the wall or ball exerted on the trunk, including friction forces. Overall, patellofemoral force and stress were greater performing the bodyweight wall squat compared with the bodyweight ball squat. Moreover, squatting with short HTWD produced anterior knee displacement beyond the toes at higher knee angles, resulting in greater patellofemoral force and stress compared with squatting with long HTWD. **Key Words:** ANTERIOR KNEE PAIN, KINETICS, SQUAT, PATELLA, REHABILITATION

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It is not currently known what patellofemoral joint force or stress magnitudes, and over what time interval, lead to patellofemoral pathology. There are numerous factors that can contribute to patellofemoral pathology, such as weakness in the quadriceps or hip external rotators, tight quadriceps, hamstrings, or iliotibial band, overuse or trauma, dysfunctional extensor mechanism, lower extremity malalignment, and excessive rear-foot pronation. Repetitive and high patellofemoral stress may cause or worsen patellofemoral pain syndrome (PFPS) and adversely affect soft tissues around the patellofemoral joint, such as infrapatellar fat pad, synovial plicae, retinacula, joint capsule, and patellofemoral ligaments (1). High patellofemoral joint force

can also increase subchondral bone stress in the patellofemoral joint (2). Because subchondral bone plate is rich in pain receptors (3), increased subchondral bone stress may result in or exacerbate PFPS (1). Patellofemoral joint stress can result in a decrease in the ability of the cartilage to absorb and distribute patellofemoral force, leading to cartilage degeneration (2). Nevertheless, better understanding of patellofemoral joint force and stress magnitudes among different weight-bearing exercises, technique variations, and functional activities may help clinicians better be able to make informed decisions regarding exercise prescription during rehabilitation.

Repetitive and high patellofemoral forces that occur during sport can result in high patellofemoral joint stress (patellofemoral force/patella contact area), which over time can lead to FPS. Squatting exercises, such as the ball and wall squat, strengthen both thigh and hip musculature and are important training and rehabilitation exercises to enhance the stability of the patellofemoral joint and improve an optimal interaction between the patella and femur during sport and activity (4–6). Knowing what force and stress magnitudes at the patellofemoral joint are generated and how they vary while using the ball and wall squat with heel-to-wall-distance (HTWD) variations may be helpful to clinicians when prescribing and progressing squatting exercises to individuals with PFPS.

Escamilla and colleagues (6) conducted the only known study involving patellofemoral force and stress during the wall squat using a 12-repetition maximum (12RM) intensity. However, there are no known studies that have examined patellofemoral force and stress during the bodyweight only wall squat and ball squat exercises, which are more common in rehabilitation settings than performing wall and ball squats with external loads. In patellofemoral rehabilitation progression, squatting exercises are initially performed with no external resistance (bodyweight only) and progressed to using external weights (dumbbells or barbells) or other external resistance, such as resistance bands, and this progression increases both thigh and hip muscle recruitment and patellofemoral force and stress (4,6,7). Moreover, both wall and ball squat exercises are commonly performed and progressed in knee rehabilitation settings using different HTWD variations. Therefore, the purpose was to quantify patellofemoral force and stress between two squat type variations (ball squat vs wall squat) and between two HTWD variations (long HTWD vs short HTWD). The hypotheses were that patellofemoral force and stress would be greater performing ball squat than wall squat during squat descent, greater in wall squat than ball squat during squat ascent, and greater with short HTWD than long HTWD.

METHODS

Subjects. Sixteen healthy participants (eight men and 8 women) without a history of patellofemoral pathology participated with a mean ($\pm$ SD) age, mass, and height of 28.9 ± 7.9 yr, 77.3 ± 6.6 kg, and 175.9 ± 2.3 cm, respectively, for male participants, and 30.6 ± 9.8 yr, 61.2 ± 6.8 kg, and 166.4 ± 8.5 cm, respectively, for female participants. Inclusion

criteria included all participants being able to perform wall and ball squats pain-free with proper technique for 12 repetitions using bodyweight and having at least 5-yr experience in performing the squatting type exercises, and exclusion criteria included not able to perform wall and ball squat exercises with varying HTWD. All participants provided written informed consent in accordance with the Institutional Review Board at California State University, Sacramento.

Exercise description. Each participant attended a pretest session 1 wk before testing and practiced performing the wall and ball squat exercises with a long HTWD and short HTWD. For the wall squat, the surface of the wall was smooth, and a towel was positioned between the wall and the subject to minimize friction as the subject moved down and up the wall. The foot position relative to the wall for the wall squat with a long HTWD (wall squat long (WSL)) and ball squat with a long HTWD (ball squat long (BSL)) was determined using an HTWD that resulted in the legs being approximately vertical at the lowest position of the squat (thighs approximately parallel with the ground) with the knees directly above the ankles (Figs. 1A, B), which is commonly recommended by clinicians and trainers. The HTWD used for the wall squat with a short HTWD (wall squat short (WSS)) was defined as one-half of the HTWD used for the WSL, whereas the HTWD used for the ball squat with a short HTWD (ball squat short (BSS)) was defined as the HTWD of BSL minus the HTWD used for WSS (Figs. 1C, D). The mean ($\pm$ SD) HTWD for male and female participants, respectively, was 43.7 ± 3.3 and 43.1 ± 2.6 cm for WSL, 21.8 ± 1.7 and 21.5 ± 1.3 cm for WSS, 93.4 ± 4.4 and 94.5 ± 3.5 cm for BSL, and 71.6 ± 3.2 and 73.0 ± 2.9 cm for BSS.

The starting position all wall and ball squat variations were standing upright with both feet slightly wider than hip width (mean inside heel to inside heel distance of 32.7 ± 7.2 cm for men and 28.9 ± 10 cm for women), feet slightly turned outward, the right foot on an AMTI force platform (Model OR6-6-2000; Advanced Mechanical Technologies, Inc.), the left foot on the ground, both knees fully extended (0° knee angle), and the trunk being either flat against the wall (wall squat) or the trunk (lower back portion) being positioned against a 65.5-cm-diameter exercise ball (ball squat). From the starting position, each participant performed a squat descent until the thighs were approximately parallel with the floor, resulting in approximately 90° – 100° of knee flexion for WSL and BSL and approximately 100° – 110° of knee flexion for WSS and BSS. Then in a continuous motion, each participant ascended upward back to the starting position. The trunk remained in approximately a vertical position for all squat variations (Figs. 1A–D). A metronome was used to help ensure the right knee flexed and extended slowly at approximately $45^\circ \cdot s^{-1}$ during the squat descent and squat ascent for all squat variations.

Data collection. Blue Sensor (Ambu Inc., Linthicum, MD) surface disposable electrodes (type M-00-S; 22 mm wide, 30 mm long) were used to collect EMG data and were placed along the longitudinal axis of each muscle in a bipolar configuration, with a center-to-center distance of 3 cm between electrodes. Before

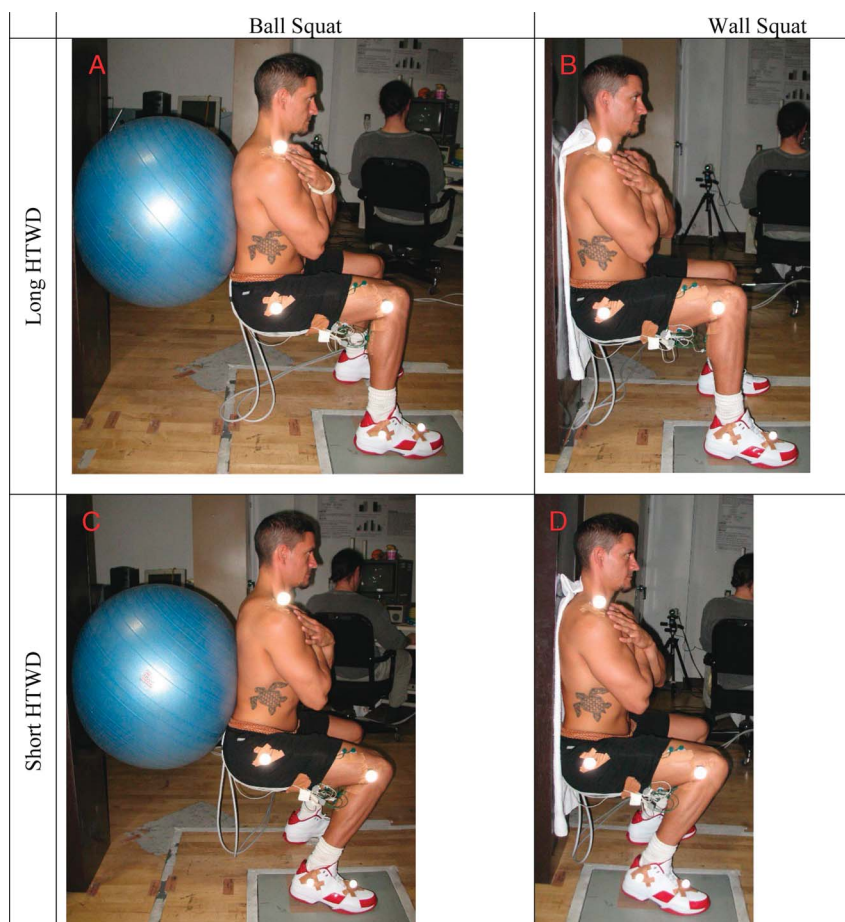


FIGURE 1—BSL HTWD (A), WSL HTWD (B), BSS HTWD (C), and WSS HTWD (D).

applying the electrodes, the skin was shaved, abraded, and cleaned with isopropyl alcohol wipes to decrease skin impedance. Using previously described locations (4–6), electrode pairs were positioned on the participant's right side for the following muscles: a) vastus lateralis, b) vastus medialis, c) rectus femoris, d) lateral hamstrings (biceps femoris), e) medial hamstrings (semimembranosus and semitendinosus), and f) gastrocnemius (middle portion between lateral and medial bellies).

For three-dimensional (3D) motion capture, spheres (3.8 cm in diameter) that were covered with reflective tape were attached to adhesives and placed over the following bony landmarks as described previously (4–6): a) third metatarsal head of the right foot, b) medial and lateral malleoli of the right leg, c) upper edges of medial and lateral tibial plateaus of the right knee, d) posterosuperior greater trochanters of the left and right femurs, and e) lateral acromion of the right shoulder.

After the electrodes and spheres were in place, the participant warmed up and practiced all exercises as needed, and then data collection commenced. An eight-camera Vicon-Peak Performance motion analysis system (Vicon-Peak Performance Technologies, Inc., Englewood, CO) was used for 60-Hz video data collection. Force data were collected at 960 Hz using an AMTI force platform (Model OR6-6-2000; Advanced Mechanical Technologies, Inc.). EMG data were collected at 960 Hz using a

Noraxon Myosystem unit (Noraxon USA, Inc., Scottsdale, AZ). EMG amplifier bandwidth frequency was 10–500 Hz with an input impedance of 20,000 k Ω , and the common-mode rejection ratio was 130 dB. Video, EMG, and force data were all electronically synchronized and simultaneously collected in a randomized order as each participant performed one set of three repetitions for WSL, BSL, WSS, and BSS.

Subsequent to completing all four squat types with their HTWD variations, EMG data were collected during maximum voluntary isometric contractions (MVIC) to normalize EMG data collected during each squat type variation, as previously described (4–6). The MVIC values for the vastus lateralis, vastus medialis, and rectus femoris were collected in a seated position at 90° knee and hip flexion during a maximum effort knee extension. The MVIC values for the medial and lateral hamstrings were collected in a seated position at 90° knee and hip flexion during a maximum effort knee flexion. The MVIC for the gastrocnemius was collected during maximum effort standing unilateral stance toe raise with the ankle positioned halfway between neutral and full plantar flexion. Two trials (each 5 s) were collected for each MVIC for each muscle in a randomized order for all three muscle groups.

Data reduction. Video images for all reflective markers were tracked and digitized in 3D space with Vicon-Peak Performance software, utilizing direct linear transformation calibration

method. The accuracy of the calibration system resulted in reflective markers that were located in 3D space with an error less than 0.3 cm. Raw position data were smoothed using a double-pass fourth-order Butterworth low-pass filter, using a cutoff frequency of 6 Hz (4–6). Joint angles, linear and angular velocities, and linear and angular accelerations were calculated utilizing appropriate kinematic equations, as previously described (4–6).

Raw EMG signals were full-waved rectified, smoothed with a 10-ms moving average window, and then linear enveloped throughout the knee range of motion for all repetitions (4–6). Subsequently, EMG data were normalized for each muscle and expressed as a percentage of each participant's highest corresponding MVIC trial. The MVIC was calculated using the highest EMG signal over a 1-s time interval during the 5-s MVIC trials, as previously described (4–6). Normalized EMG data for all three repetitions (trials) were averaged at corresponding knee angles between 0° and 100° with 0° defining full knee extension, 0° and 100° defining the squat descent, and 100° and 0° defining the squat ascent. EMG data were used to calculate patellofemoral force and stress in a biomechanical knee model explained in Appendix 1 (see Supplemental

Digital Content, <http://links.lww.com/MSS/C847>) and were not analyzed separately.

Data analysis. A repeated-measures two-way ANOVA was used for each 10° knee angle (from 10° to 100°) during the squat descent and each 10° knee angle (from 100° to 10°) during the squat ascent to assess the effects of squat type (ball vs wall squat) and HTWD (long HTWD vs short HTWD) on patellofemoral compressive force and stress. A repeated-measures three-way ANOVA was also used separately for the knee flexing (descending) phase and the knee extending (ascending) phase to test the overall effect of knee angles during these two phases on patellofemoral force and stress. The level of significance was set at $P < 0.01$. Bonferroni t -tests were used to assess pairwise comparisons among the squat conditions.

RESULTS

Descriptive data for calculated patellofemoral joint force and stress for each squat type and HTWD condition are provided in Figures 2A and 3B. Visual observation of the data suggests that patellofemoral joint force and stress generally

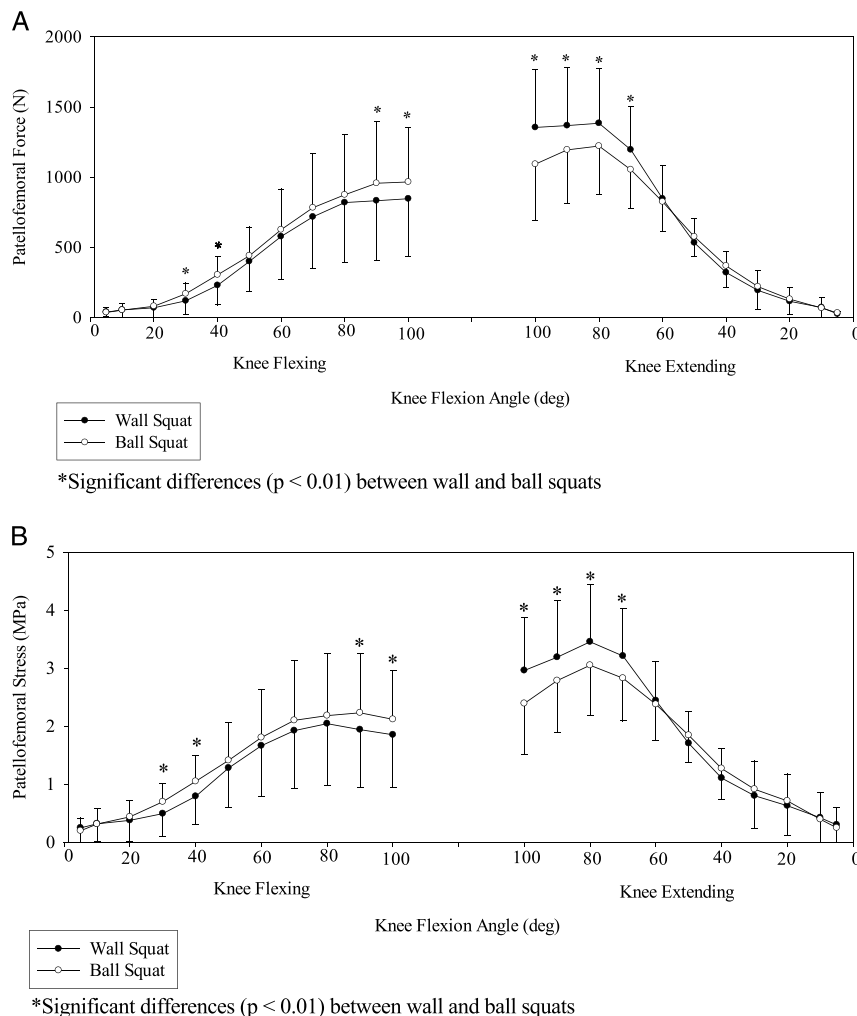


FIGURE 2—A, Mean (SD) patellofemoral compressive force between ball squat and wall squat collapsed across HTWD. B, Mean (SD) patellofemoral stress between ball squat and wall squat collapsed across HTWD.

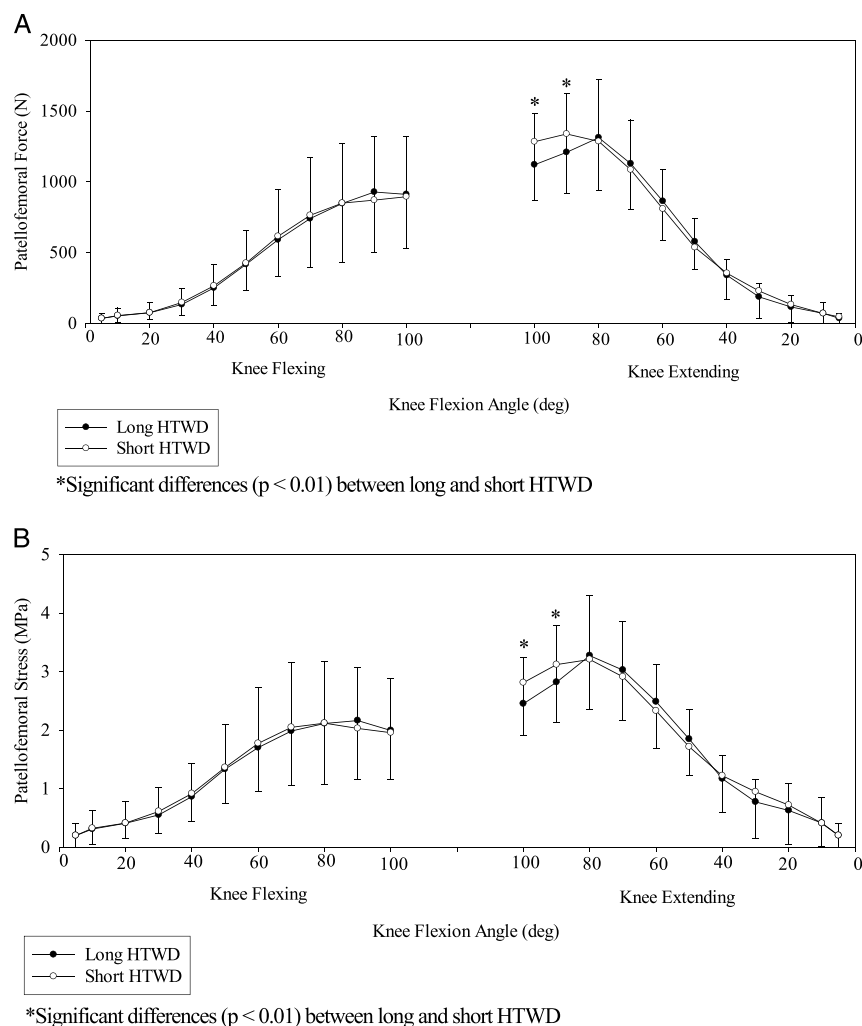


FIGURE 3—A, Mean (SD) patellofemoral compressive force between squatting with long and short HTWD collapsed across squat type. B, Mean (SD) patellofemoral stress between squatting with long and short HTWD collapsed across squat type.

increased progressively as knee flexion increased during the descent phase and decreased progressively as knee flexion decreased during the ascent phase. Moreover, for a given knee angle, patellofemoral joint force and stress were generally slightly greater during the ascent phases compared with the descent phases.

Table 1 and Figures 2A–B provide patellofemoral joint force and stress values between the two squat type conditions (ball squat vs wall squat) collapsed across the two HTWD conditions. The P values shown in Table 1 for the squat type conditions represent the main effects of the ANOVA. When collapsed across long and short HTWD, patellofemoral joint force and stress were significantly greater in ball squat than wall squat at 30°, 40°, 90°, and 100° knee angles (0° = full knee extension) during squat descent, but significantly greater in the wall squat than the ball squat at 100°, 90°, 80°, and 70° knee angles.

Table 1 and Figures 3A–B show patellofemoral joint force and stress values between the two HTWD conditions (long HTWD vs short HTWD) collapsed across the two squat conditions. The P values shown in Table 1 for the two HTWD

conditions represent the main effects of the ANOVA. When collapsed across ball and wall squats, patellofemoral joint force and stress were significantly greater in short HTWD compared with long HTWD at 100° and 90° knee angles during squat ascent. Regarding effect size, for squat type variations, there were generally large size effects in patellofemoral loading throughout the squat descent and ascent, and for HTWD variations, there were generally small to large size effects in patellofemoral loading throughout the squat descent and ascent (Table 1). There were no significant interactions between squat type and HTWD variations throughout the squat descent and ascent.

The repeated-measures three-way ANOVA revealed that there were significant differences in all knee angles collectively between two squat types for both squat descent ($P = 0.005$) and ascent ($P = 0.004$; Table 1), and no significant differences in all knee angles collectively between HTWD variations for both squat descent ($P = 0.773$) and ascent ($P = 0.097$; Table 1). There were significant influences of knee angle on patellofemoral loading with overall trend of the patellofemoral loading

TABLE 1. Mean ($\pm$ SD) patellofemoral joint force (N) values while performing the ball squat and wall squat with HTWD variations.

	Squat Type Variations		<i>P</i>	Effect Size	HTWD Variations		<i>P</i>	Effect Size
	Ball Squat	Wall Squat		η_p^2	Long	Short		η_p^2
Knee angles for squat descent phase								
10°	54 ± 46	55 ± 53	0.929	0.021	53 ± 46	55 ± 53	0.851	0.031
20°	81 ± 51	70 ± 66	0.232	0.091	75 ± 47	76 ± 69	0.963	0.015
30°	170 ± 76	120 ± 95	0.002*	0.462	133 ± 76	148 ± 98	0.282	0.080
40°	305 ± 130	230 ± 140	0.002*	0.447	251 ± 124	266 ± 149	0.393	0.053
50°	441 ± 203	400 ± 212	0.143	0.148	417 ± 182	426 ± 232	0.660	0.034
60°	627 ± 289	578 ± 305	0.192	0.151	591 ± 260	615 ± 331	0.473	0.053
70°	784 ± 387	718 ± 372	0.075	0.144	740 ± 349	764 ± 410	0.596	0.037
80°	876 ± 430	820 ± 429	0.204	0.101	848 ± 416	850 ± 424	0.795	0.026
90°	958 ± 438	834 ± 430	0.003*	0.285	928 ± 430	871 ± 446	0.377	0.047
100°	967 ± 389	847 ± 412	0.003*	0.232	912 ± 380	895 ± 424	0.900	0.027
All knee angles collectively			0.005*				0.773	
Knee angles for squat ascent phase								
100°	1194 ± 403	1355 ± 414	<0.001*	0.211	1120 ± 362	1283 ± 412	0.009*	0.260
90°	1195 ± 382	1368 ± 416	<0.001*	0.297	1208 ± 418	1338 ± 424	0.008*	0.185
80°	1223 ± 348	1385 ± 393	0.004*	0.348	1311 ± 409	1285 ± 345	0.639	0.019
70°	1054 ± 274	1196 ± 306	0.003*	0.201	1128 ± 308	1085 ± 280	0.331	0.027
60°	825 ± 214	847 ± 235	0.546	0.078	863 ± 222	808 ± 224	0.178	0.136
50°	576 ± 144	533 ± 170	0.013	0.273	577 ± 160	536 ± 154	0.207	0.156
40°	368 ± 154	321 ± 149	0.014	0.540	339 ± 115	354 ± 184	0.639	0.108
30°	221 ± 163	194 ± 144	0.090	0.314	187 ± 94	228 ± 194	0.230	0.464
20°	131 ± 111	116 ± 100	0.176	0.152	115 ± 84	132 ± 123	0.224	0.204
10°	68 ± 71	72 ± 73	0.973	0.025	70 ± 76	69 ± 68	0.802	0.027
All knee angles collectively			0.004*				0.097	

The mean values given for the two squat type conditions (ball squat and wall squat) were collapsed across the two HTWD conditions (long HTWD and short HTWD), whereas the mean values given for the two HTWD conditions were collapsed across the two squat type conditions. The *P* values shown for squat type conditions and HTWD conditions represent the main effects from the repeated-measure two-way ANOVA. The *P* values shown for all knee angles collectively represent the main effect from the repeated-measures three-way ANOVA.

*Significant difference ($P < 0.01$) between squat type conditions or between HTWD conditions.

η_p^2 , partial eta squared.

increasing with knee angle for both squat descent ($P < 0.001$) and ascent ($P < 0.001$; Figs. 2A, 3B).

DISCUSSION

This is the first known study that examined the effects of performing wall and ball squats with long and short HTWD on patellofemoral force and stress. These findings provide insight into how the patellofemoral joint can be loaded and progressed in PFPS rehabilitation using ball and wall squats with long and short HTWD. Therefore, squatting progression using lower to higher patellofemoral loading over time during PFPS rehabilitation may proceed as follows: 1) ball or wall “mini” squats at lower more functional knee angles (0°–50°) with either long or short HTWD, 2) BSL at 0°–100° knee angles, 3) WSL at 0°–100° knee angles, 4) BSS at 0°–100° knee angles, and 5) WSS at 0°–100° knee angles.

The bodyweight squat, as in the current study, is more appropriate earlier in PFPS rehabilitation, starting at lower knee angles (0°–50°) with lower patellofemoral loading and progressing to higher knee angles (50°–100°) with higher patellofemoral loading. Kernozek et al. (8) also examined the bodyweight squat using two techniques, one squatting with knees past toes (SPT), similar to WSS and BSS in the current study, and one squatting with knees behind toes (SBT), similar to WSL and BSL in the current study. They reported mean patellofemoral force and stress of approximately 1450 N and 3.4 MPa (both similar to the 1200–1400 N and 3.05–3.45 MPa in the current study), respectively, for SBT, and approximately 1950 N and 4.2 MPa, respectively, for SPT. Therefore, like the current study, these authors reported greater

patellofemoral force and stress when knees moved beyond toes (similar to our short HTWD) compared with when knees stayed over the feet (similar to our long HTWD). Wallace et al. (9) reported peak patellofemoral joint force and stress magnitudes of approximately 1700 N (also similar to our force magnitudes) and 9.3 MPa (2–3 times greater than our magnitudes), respectively, during the bodyweight squat at 90° knee angle. Almonroeder et al. (10) reported a mean patellofemoral joint force and stress magnitudes of approximately 2300 N and 11 MPa, respectively, while performing the bodyweight squat.

With patellofemoral loading now known when performing the bodyweight ball and wall squats, the results from the current study can be compared with patellofemoral loading from other types of squatting and with varying intensities to help clinicians progress a patient with PFPS. Escamilla et al. (4,5,7) reported a peak patellofemoral joint force and stress magnitudes of 4500–4700 N and 11–12 MPa, respectively, at 90° knee angle while performing the 12RM barbell squat, approximately 3–4 times greater than peak magnitudes in the current study. Escamilla et al. (6) also reported peak patellofemoral joint force and stress magnitudes of approximately 3500 N and 9 MPa, respectively, between 70° and 80° knee angle while performing the 12RM wall squat and one leg squat, approximately 2–3 times greater than peak magnitudes in the current study. Wallace et al. (9) reported peak patellofemoral joint force and stress magnitudes of approximately 2400 N and 13 MPa, respectively, while performing the barbell squat with a 35% bodyweight external load, approximately 2–4 times greater than peak magnitudes from the current study. Based on patellofemoral loading when performing these different squat exercises with varying intensities, when the goal is to initially

minimize patellofemoral loading (e.g., early phase of PFPS rehabilitation) and later progress patellofemoral loading (e.g., later phase of PFPS rehabilitation), the order of squatting progression could be to initially perform BSL and gradually progressing to BSS, WSL, and WSS, followed by a lower-intensity barbell squat (e.g., one-third bodyweight external load), a higher-intensity wall squat (e.g., 12RM external load), and a higher-intensity barbell squat (e.g., 12RM external load).

Peak patellofemoral joint force and stress magnitudes from the current study are similar, less than, or greater than peak magnitudes in many functional activities. Peak patellofemoral joint force and stress in healthy participants performing fast walking are approximately 900 N and 3.13 MPa, respectively (11), which are similar to (although slightly less than) peak magnitudes in the current study. Peak patellofemoral joint force and stress magnitudes in healthy participants going up and down stairs are approximately 2500 N and 7 MPa, respectively (11), which is approximately twice the peak magnitudes in the current study. Based on patellofemoral loading in the bodyweight ball and wall squats with varying HTWD and patellofemoral loading in these different functional activities with varying intensities, when the goal is to initially minimize and later progress patellofemoral loading, the order of progression when integrating functional activities with the bodyweight ball and wall squats with HTWD variations could be to initially perform slow walking; progress to fast walking, progress to BSL and gradually progressing to BSS, WSL, and WSS; and finally progress to going up and down stairs.

During any squat exercise, there are always at least two external forces acting on the body, which are 1) vertical upward directed ground reaction force acting on the feet and 2) vertical downward directed gravitational force acting at the body's center of mass. During the wall and ball squat, there is a third external force acting on the body, which is the force the wall or ball exerts on the trunk. The third external force can be resolved into two force components—one force acting perpendicular to the trunk called the “normal” force (F_N), and the other component force acting parallel with the trunk called the friction force (F_{fr}), which is a force that always opposes motion or impending motion by acting in the opposite direction of the motion. Because the body is sliding up and down the wall, the friction force during the wall squat is defined as kinetic (or sliding) friction $F_{\text{kinetic-fr}} = \mu_k F_N$, where μ_k is the coefficient of kinetic friction. The value for μ_k is dependent on the “roughness” of the surfaces sliding relative to each other, with smoother surfaces decreasing μ_k and rougher surfaces increasing μ_k . During the ball squat, because the ball rolls both relative to the body and relative to the wall, instead of a sliding friction force, there is a rolling friction force $F_{\text{rolling-fr}} = \mu_r F_N$, where μ_r is the coefficient of rolling friction. Because μ_r is always much less than μ_k , and because F_N will be similar between WSL and BSL and between WSS and BSS (given the heel-to-wall and heel-to-ball distances are similar to each other), the rolling friction force during the ball squat will be much less than the kinetic friction force during the wall squat. This implies that friction force was greater in

WSL than BSL and greater in WSS than BSS for both the squat descent and ascent.

During the squat descent, the friction force acts vertically upward, subtracting from the vertical downward acting gravitational force, thus decreasing the net knee flexor torque generated from the external forces acting on the body and decreasing the net knee extensor torque from quadriceps force needed. This implies that less patellofemoral force generated in wall squat than ball squat, because patellofemoral force decreases as quadriceps force decreases and increases as quadriceps force increases, which is reflective in less overall quadriceps activity in the wall squat than the ball squat during the squat descent (see Supplemental Table 1, Supplemental Digital Content, Mean normalized EMG values for the quadriceps, hamstrings, and gastrocnemius muscles while performing the ball squat and wall squat collapsed across the two HTWD conditions, <http://links.lww.com/MSS/C847>). The larger friction force in wall squat than ball squat makes it easier to perform wall squat during descent but harder to perform wall squat during ascent when friction force acts vertically downward. This downward friction force is summated to gravitational force, increasing the net knee flexor torque generated from external forces, and increasing the net knee extensor torque by greater quadriceps force generated (thus increasing patellofemoral force), which is reflective in greater overall quadriceps activity in the wall squat than the ball squat during the squat ascent (see Supplemental Table 1, Supplemental Digital Content, <http://links.lww.com/MSS/C847>). In a clinical application, when the goal is to initially minimize and later progress patellofemoral loading, the order of progression would be to initially perform only the descent phase of WSL and WSS, progressing to only the descent phase of the BSL and WSS, then progressing to the ascent phase of the BSL, BSS, WSL, and finally WSS.

Because friction force is defined as $F_{\text{fr}} = \mu F_{\text{N}}$, friction force F_{fr} is directly proportional to the normal force F_{N} . Because F_{N} increases progressively as the feet move farther away for the wall, F_{fr} also increases progressively. Therefore, both F_{N} and F_{fr} will be greater in WSL and BSL (long HTWD) than in WSS and BSS (short HTWD). Therefore, compared with a short HTWD, one may expect wall and ball squats with a long HTWD to generate less patellofemoral force during descent (when the higher friction force makes it easier) and more patellofemoral force during ascent (when the higher friction force makes it harder). However, F_{fr} is not the only factor that influences patellofemoral loading. For example, other squat studies (6,8) have demonstrated that when the knees move forward beyond the toes, such as what occurred during WSS and BSS (Figs. 1C, D), patellofemoral force increases and is greater compared with when the knees do not progress beyond toes and stay over feet, such as what occurred during WSL and BSL (Figs. 1A, B). This implies that using a long HTWD may increase patellofemoral loading during the squat ascent from an increase in F_{fr} , but using a short HTWD may also increase patellofemoral loading during the squat ascent secondary to increased forward knee motion beyond the toes. Therefore, because both a long HTWD and short HTWD seem to increase

patellofemoral loading, this may help explain why patellofemoral forces were generally not significantly different between squatting with a long and short HTWD. The exception was near maximum knee flexion (90° and 100° knee angles) during ascent, when the knees had their greatest forward knee displacement beyond toes during short HTWD, which may affect patellofemoral loading to a greater extent compared with F_{fr} . At these maximum knee flexion angles, squatting with a short HTWD produced significantly greater patellofemoral force and stress compared with squatting with a long HTWD.

From qualitative analyses, graph curves (Figs. 2A, 3B) for patellofemoral joint force and stress were similar in shape to each other because of near proportional increases in patellar contact areas and patellofemoral joint forces with increased knee flexion. An exception was with higher knee angles between 70° and 100°, in which patellofemoral joint stress started plateauing or decreased. This happened because, although patellar contact area increased near linearly between 70° and 100°, patellofemoral joint force did not increase proportionally, but instead began to plateau or decrease around 70°–80° in the current study as well as in other squat studies in the literature (4–6). Thus, the risk of injury to the patellofemoral joint may not increase with knee angles between 70° and 100° or greater because of the similar magnitudes in patellofemoral joint stress during these knee angles, with a benefit of greater hamstrings, quadriceps, and gluteus maximus activity during training at 70°–100° knee angles or higher compared with training at lower knee angles between 0° and 60°, which is more quadriceps dominant (4–6).

There are limitations to the current study. First, knee kinematic magnetic resonance imaging (MRI) data have demonstrated that during weight-bearing squat, the femur moves and rotates underneath a relatively stationary patella, and too much femoral rotation may increase patellofemoral joint stress on the contralateral patellar facets (12). Unfortunately, knee kinematic MRI data do not currently exist while performing wall or ball squat exercises through functional ranges of motion. Therefore, it is not known how much the femur rotates and how this rotation varies among healthy individuals compared with those with pathologies. Second, all biomechanical models have modeling limitations (see Supplemental Digital Content for limitations of the biomechanical model, <http://links.lww.com/MSS/C847>). Third, patellofemoral joint stress magnitudes were calculated using patellar contact area values from MRI data from the literature and not measured directly

for each subject. However, contact areas used from the literature were determined while performing loaded weight-bearing exercise in healthy male and female participants, similar to the current study. Moreover, the near linear and direct relationship between knee angle and contact area is similar among studies (2,13,14). This implies that patellofemoral joint stress curve patterns in Figures 2B and 3B using contact areas from the literature are similar to patellofemoral joint stress curve patterns for contact areas measured directly from MRI. Patellofemoral joint stress patterns are important for clinicians to understand to help determine what knee range of motions patellofemoral joint stress goes up or down, which can exacerbate PFPS. Lastly, the focus of the current study was to compare right knee patellofemoral force/stress between the ball squat and wall squat, so we only chose only the right leg to use in the knee model to assess this using only one force platform. The focus was not to compare patellofemoral force/stress between left and right knees and assess symmetry between left and right sides of the body. This should be the focus of a future study.

CONCLUSIONS

Patellofemoral joint loading changed according to variations in both squat type and HTWD variations. These findings were in part due to differences in how these two squat exercises were performed, with wall squats involving sliding up and down the wall with kinetic friction helping or resisting the movements, and ball squats involving rolling up and down the wall with rolling friction (which is less than kinetic friction) helping or resisting the movements, and friction forces in both the wall and ball squats being greater with long HTWD compared with short HTWD because of a greater “normal” force with long HTWD. Overall, patellofemoral force and stress were greater performing the bodyweight wall squat compared with the bodyweight ball squat. Moreover, squatting with short HTWD produced anterior knee displacement beyond the toes at higher knee angles, resulting in greater patellofemoral force and stress compared with squatting with long HTWD.

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